

**The University of Texas at Austin
Department of Chemical Engineering
ChE 350: Chemical Engineering Materials
UNIQUE # 17065; Fall 2026**

Instructor: Brian A. Korgel

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Course Mode: in-person. Contact Prof. Korgel if you have any questions about attendance.

MEETING TIMES: Lecture: M/W/F, 11:00 am-12:00 pm (noon)—CPE 2.214
Recitation: F, 8:00 am-9:00 am—CPE 2.214

This course provides an introduction to materials science and engineering. Materials science focuses on understanding the structure-property relationships of solids and condensed phases of matter; i.e., how materials respond to applied stimuli, such as light, electricity, mechanical forces, and magnetic fields. By understanding the fundamental relationships between atomic structure and properties, materials can be designed to elicit a desired set of properties and used in a wide range of important applications.

Course Outline:

Chemical Bonding and Materials Structure; Crystal structure and Defects; Polymers; Mechanical properties; Electrical properties; Optical properties; Magnetic properties; Thermal Properties; Corrosion; Biomaterials & Composites; Nanomaterials

Texts: (1) W. D. Callister & D. G. Rethwisch, Materials Science and Engineering: An Introduction, 9th Edition (John Wiley & Sons, New York, NY); ISBN: 978-1-118-32457-8 **or** 10th Edition; ISBN: 978-1-119-40549-8
(2) Mark Miodownik, Stuff Matters: Exploring the Marvelous Materials that Shape our Man-Made World (Houghton Mifflin Harcourt Publishing Company, New York, NY); ISBN: 978-0-544-48394-1

Prerequisites: Chemistry 353 with a grade of at least C-. Chemical Engineering Major or consent of instructor.

Course Requirements: The coursework will consist of reading, homework, three exams, and one final exam. ***Late homework is not accepted and will not be graded.***

Grading:	Homework (10 assignments, due <u>Mondays</u> before the start of class)	15%
	Exams (Mondays, 9/23, 10/30, 11/11)	60%
	Final exam (Monday 12/11; 8:00 am-10:00 am)	25%

Grades are not curved: >88% (A); 85-88 (A-); 82-85 (B+); 73-82% (B); 70-73 (B-); 67-70 (C+); 58-67% (C); 55-58 (C-); <55% (D)

The University of Texas at Austin provides, upon request, appropriate academic adjustments for qualified students with disabilities. For more information, please contact the Office of the Dean of Students at 471-6259, 471-4641 TTY.

Homework. Students will scan and upload their homework assignments to canvas. Make sure to leave enough time to do this before the start of class. All homeworks go through a standard plagiarism and AI check.

Exams. Exams will be taken in person in CPE 2.214 during the regularly scheduled class time. Exams will be closed book, closed notes, no calculator.

The University of Texas Honor Code. The core values of the University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the University is expected to uphold these values through integrity, honesty, trust, fairness, and respect towards peers and community.

COVID guidelines. UT encourages all students, faculty and staff members to get vaccinated. COVID-19 vaccines are our best tool to help us #ProtectTexasTogether. Information is provided at <https://protect.utexas.edu/vaccines-testing/>. If you feel sick, stay home.

Land Acknowledgment

(I) We would like to acknowledge that we are meeting on Indigenous lands of Turtle Island, the ancestral name for what now is called North America. Moreover, (I) We would like to acknowledge the Alabama-Coushatta, Caddo, Carrizo/Comecrudo, Coahuiltecan, Comanche, Kickapoo, Lipan Apache, Tonkawa and Ysleta Del Sur Pueblo, and all the American Indian and Indigenous Peoples and communities who have been or have become a part of these lands and territories in Texas.